

A COMPARISON OF EXACT AND ASYMPTOTIC PROPERTIES OF TESTS OF POWER ONE FOR THE LOCATION PARAMETER OF AN EXPONENTIAL DISTRIBUTION

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SUMMARY

In a preceding paper (Swanepoel and Geertsema (1976(a)) tests of power one for $H_0: \theta \leq \theta_0$ against $H_1: \theta > \theta_0$ are considered in the case of an exponential density $f_\theta(x) = \exp \{-(x - \theta)\} I(x \geq \theta)$. It is proved that a test which stops and rejects H_0 at the first n for which $M_n \geq c_n$ (where M_n is the minimum of the first n observations and $\{c_n\}$ is a decreasing sequence of constants) is asymptotically as $\theta \rightarrow \theta_0$ optimal, where $c_n \sim n^{-1} \log_2 n$. (Write $\log_2 = \log \log$, etc.)

In the present article we derive the exact distribution of the sample size N of tests of the above type. This enables us to compare the asymptotic expressions for $E_\theta N$ as $\theta \rightarrow 0$ (we take $\theta_0 = 0$ without loss of generality) and as $\alpha \rightarrow 0$ with the exact values. We also compare an asymptotically optimal procedure with a competitor with $c_n \sim n^{-1} \log n$ and the latter is shown to be more efficient for θ even quite close to 0. We are not aware of similar existing exact results on power one tests.

1. The exact distribution of sample size

Theorem 1.1

Let $\{c_n, n \geq m\}$ be a nonincreasing sequence of constants and $M_n = \min(X_1, \dots, X_n)$, $n = m, m+1, \dots$, where $X_1, X_2, \dots$ are independent observations with common distribution function F . Let N_m be the first integer $n \geq m$ such that $M_n \geq c_n$ and let $L_n^m = P(N_m = n)$, $P_n^m = P(N_m > n)$ and $x_n = 1 - F(c_n)$. Then

$$(1.1) \quad L_n^m = x_n^n - \sum_{j=m}^{n-1} x_n^{n-j} L_j^m$$

$$(1.2) \quad P_n^m = x_n^{-1} (1 - x_n) \sum_{j=m}^{n-1} x_n^{n-j} P_j^m + x_n^{n-m} (1 - x_n^m)$$

Proof:

We derive (1.1). The proof of (1.2) is analogous or (1.2) may be derived from (1.1).

$$\begin{aligned}
L_n^m &= P(M_m < c_m, \dots, M_{n-1} < c_{n-1}, M_n \geq c_n) \\
&= P(M_m < c_m, \dots, c_n \leq M_{n-1} < c_{n-1}) (1 - F(c_n)) \\
&= P(M_m < c_m, \dots, M_{n-2} < c_{n-2}, M_{n-1} \geq c_n) (1 - F(c_n)) - L_{n-1}^m (1 - F(c_n)) \\
&= P(M_m < c_m, \dots, c_n \leq M_{n-2} < c_{n-2}) (1 - F(c_n))^2 - L_{n-1}^m (1 - F(c_n)) \\
&= P(M_m < c_m, \dots, M_{n-3} < c_{n-3}, M_{n-2} \geq c_n) - L_{n-2}^m (1 - F(c_n))^2 \\
&\quad - L_{n-1}^m (1 - F(c_n)) \\
&\dots \\
&= P(M_m < c_m, M_{m+1} \geq c_n) (1 - F(c_n))^{n-m-1} - L_{m+1}^m (1 - F(c_n))^{n-m-1} - \dots \\
&\quad - L_{n-2}^m (1 - F(c_n))^2 - L_{n-1}^m (1 - F(c_n))
\end{aligned}$$

Then (1.1) follows, since

$$P(M_m < c_m, M_{m+1} \geq c_n) = (1 - F(c_n))^{m+1} - (1 - F(c_n)) L_m^m$$

Remark

Using Theorem 1.1, the distribution of N may be calculated recursively.

2. Construction and comparison of tests

From (1.1) we obtain

$$\begin{aligned}
L_n^m &= x_n^n - x_n L_{n-1}^m - \sum_{j=m}^{n-2} x_n^{n-j} L_j^m \\
&= x_n^n - x_n x_{n-1}^{n-1} - \sum_{j=m}^{n-2} L_j^m x_n (x_n^{n-j-1} - x_{n-1}^{n-j-1})
\end{aligned}$$

$$(2.1) \leq x_n^n - x_n x_{n-1}^{n-1} \text{ since } c_n \text{ is nonincreasing.}$$

Since $P(N < \infty) = \sum_{n=m}^{\infty} L_n^m$, this allows us the construction of tests of power one, such that $P_0(N < \infty) \leq \alpha$ for given $0 < \alpha < 1$. For example, if $F(x) = 1 - \exp\{-x\}$, $x \geq 0$, and

$$(2.2) \quad c_n = n^{-1}(\log_2 n + 3 \log_3 n + \log c),$$

we obtain from (2.1)

$$L_n^m < x_n^n y_n \text{ where}$$

$$y_n = 1 - \exp \{-(n-1)(c_{n-1} - c_n)\}$$

$$\leq (n-1)(c_{n-1} - c_n) \text{ for } n \text{ large}$$

$$\leq n^{-1}(\log_2 n + 3\log_3 n + \log c)$$

$$\text{while } x_n^n = \exp \{-nc_n\}$$

$$= (\text{clogn}(\log_2 n)^3)^{-1}$$

The sequence $\{c_n\}$ is decreasing for $n \geq 19$ and hence we choose $m = 20$. Using integral bounds for series of decreasing terms, we thus obtain

$$\sum_{n=20}^{\infty} L_n^m \leq c^{-1} (1,668184 + 0,428734 \log c)$$

This bound is 0,05 for $c = 69,8$ and 0,01 for $c = 428$.

Referring to Robbins and Siegmund (1972) it may be seen (cf. their cor. 1) that our $\{c_n\}$ of (2.2) is of as small an order as can be obtained. Let N_1 be the stopping variable of the procedure with sequence $\{c_n\}$. Exact calculations using (1.2) show that $0,033 < P_0(N_1 < \infty) \leq 0,050$ if $c = 69,8$. We used as lower bound $P_0(N_1 \leq 6000) = 0,033$. The computer time involved did not allow calculations of $P_0(N_1 \leq n)$ for larger values of n .

For purposes of comparison, let $c'_n = n^{-1}(\log n + \log d)$, $n \geq 20$, then the procedure with this sequence and $d = 4,74$ may be similarly shown to have level 5%, while $d = 34$ gives a procedure with level 1%. Exact calculations show

$$0,045 = P_0(N_2 \leq 500) < P_0(N_2 < \infty) \leq 0,050,$$

where N_2 is the stopping variable of the procedure with sequence $\{c'_n\}$.

As in Swanepoel and Geertsema (1976a) it follows that

$$(2.3) \quad E_{\theta} N_1 \sim P_0(N_1 = \infty) \theta^{-1} \log_2 \theta^{-1} \text{ and}$$

$$E_{\theta} N_2 \sim P_0(N_2 = \infty) \theta^{-1} \log \theta^{-1} \text{ as } \theta \downarrow 0$$

which shows that $E_{\theta}N_2$ is of a higher order compared with $E_{\theta}N_1$ for $\theta \downarrow 0$. However, Table 2.1 shows that the expression (2.3) cannot serve as even a crude approximation for quite small values of θ . This table gives values of the asymptotic expression (2.3), denoted by $\nu(\theta)$ for various values of θ , chosen in such a way that $\nu(\theta)$ increases by an approximately constant amount, $\alpha = 0,05$ and $\alpha = 0,01$. (In the computation of $\nu(\theta)$ we used $P_0(N_1 = \infty) = 1 - \alpha$.) The table also gives exact values of $E_{\theta}N_1$ and $E_{\theta}N_2$ for $\alpha = 0,05$ and $\alpha = 0,01$.

TABLE 2.1

α	θ	0,100	0,060	0,045	0,037	0,032	0,028	0,025	0,022	0,020	0,018	0,016
0,05	$\nu(\theta)$	0,6	7,9	16,4	23,9	30,6	43,3	49,6	57,8	64,8	73,4	84,3
0,05	$E_{\theta}N_1$	54,0	95,3	130,8	162,1	189,9	219,5	248,1	284,9	315,6	353,5	401,1
0,05	$E_{\theta}N_2$	41,7	76,0	107,4	135,8	161,7	189,8	217,4	253,3	283,8	321,7	370,1
0,01	$\nu(\theta)$	0,6	8,2	17,1	24,9	31,9	45,1	51,7	60,2	67,5	76,5	87,8
0,01	$E_{\theta}N_1$	74,9	130,3	177,5	218,7	255,3	294,2	331,8	379,8	420,0	469,4	531,3
0,01	$E_{\theta}N_2$	65,0	117,1	163,1	204,3	241,3	281,2	320,0	370,3	412,8	465,3	531,9

Since the asymptotic expression as $\theta \downarrow 0$ does not give a description of the behaviour of the expected sample size for the values of θ considered, it is not surprising to see that the asymptotic optimality of the procedure based on N_1 does not imply optimality for the values of θ which are considered. It is clear from Table 2.1 that the procedure based on N_2 is to be preferred for these θ values. This serves as motivation to study, in the next sections, a procedure similar to that based on N_2 (i.e. with a sequence $c_n \sim n^{-1} \log n$) and with $m = 3$.

3. A procedure and some properties

The following martingale theorem will be used to construct a procedure.

Theorem 3.1

If $\{f_n(x)\}$ is a sequence of real-valued differentiable functions satisfying

$$(i) \quad f'_{n+1}(x) = [1 - F(x)]^{-1} f'_n(x) \text{ for all } x, n \geq 1$$

$$(ii) \quad f_{n+1}(-\infty) = f_n(-\infty), n \geq 1$$

then $\{f_n(M_n)\}$ is a martingale sequence.

Proof:

Since $M_{n+1} = \min(M_n, X_{n+1})$,

$$\begin{aligned}
 & E(f_{n+1}(M_{n+1}) | X_n, \dots, X_1) \\
 &= E(f_{n+1}(M_n)I(X_{n+1} \geq M_n) | X_n, \dots, X_1) + E(f_{n+1}(X_{n+1})I(X_{n+1} < M_n) | X_n, \dots, X_1) \\
 &= f_{n+1}(M_n)(1 - F(M_n)) + \int_{-\infty}^{M_n} f_{n+1}(x)dF(x) \\
 &= f_n(M_n), \text{ using the assumptions (i) and (ii).}
 \end{aligned}$$

Remark

The functions

$$f_n(x) = (n-2)^{-1}[(1-F(x))^{-(n-2)} - 1], n \geq 3$$

satisfy the conditions of Theorem 3.1 if F is differentiable.

If $F(x) = 1 - \exp \{-x\}$, $x \geq 0$, and using a well-known martingale inequality, this choice of $f_n(x)$ gives

$$(3.1) \quad P(M_n \geq c_n \text{ for some } n \geq 3) \leq \alpha$$

where

$$(3.2) \quad c_n = (n-2)^{-1} \log(1 + (n-2)/2\alpha)$$

The power one test based on this sequence $\{c_n\}$ will be the subject of the remainder of the paper. Exact computations based on (1.2) show that the inequality in (3.1) is practically an equality, e.g. if $\alpha = 0,050$ we find:

$$0,049 = P(M_n \geq c_n \text{ for some } 3 \leq n \leq 5\,000)$$

$$\leq P(M_n \geq c_n \text{ for some } n \geq 3) \leq 0,050$$

Since it was seen in Section 2 that asymptotic theory as $\theta \downarrow 0$ gives poor approximations to expected sample size, we now turn to asymptotic theory as $\alpha \rightarrow 0$.

Theorem 3.2

Let N be the first $n \geq 3$ such that $M_n \geq c_n$, where c_n is defined in (3.2). Then

$$(3.3) \quad \lim_{\alpha \rightarrow 0} E_\theta N / N(\alpha, \theta) = 1 \text{ for } \theta > 0$$

where $\lambda(\alpha, \theta) = \theta^{-1} \log \alpha^{-1}$.

Proof:

From Lemma 1 of Chow and Robbins (1965), obtain

$$(3.4) \quad \lim_{\alpha \rightarrow 0} N/\lambda(\alpha, \theta) = 1 \text{ a.s.}$$

To complete the proof, it is sufficient to prove the uniform integrability of $\{N/\lambda(\alpha, \theta)\}_{\alpha > 0}$. According to Bickel and Yahav (1968) this follows if

$$\sum_{n=n_0}^{\infty} \sup_{0 < \alpha < \alpha_0} P_{\theta}(N/\lambda(\alpha, \theta) > n) < \infty$$

for some n_0 and some α_0 independent of n . Let $r = [n\lambda(\alpha, \theta)]$. Then

$$P_{\theta}(N/\lambda(\alpha, \theta) > n) \leq P_{\theta}(M_r < c_r) = 0 \text{ if } c_r < \theta.$$

But there is an $\alpha_0 > 0$ (and independent of n) such that for $\alpha < \alpha_0$

$$c_r \leq (n-1)^{-1}(1 + \theta \log n)$$

Hence $c_r < \theta$ for $\alpha < \alpha_0$ and $n \geq n_0$, for some n_0 .

Table 3.1 gives exact values of $E_{\theta}N$ for various θ and α , calculated by means of (1.2). The α -values are chosen in such a way that $\log \alpha^{-1}$ increases by approximately equal amounts. Between parentheses the corresponding values of $E_{\theta}N - \rho(\alpha, \theta)$ are given where

$$\rho(\alpha, \theta) = \theta^{-1}(\log \alpha^{-1} + \log_2 \alpha^{-1}).$$

Note that $\rho(\alpha, \theta) \sim \lambda(\alpha, \theta)$ as $\alpha \rightarrow 0$. Table 3.1 indicates that

$$(3.5) \quad E_{\theta}N \leq \rho(\alpha, \theta) + A_{\theta}$$

where A_{θ} does not depend on α .

4. The asymptotic distribution as $\alpha \rightarrow 0$ of the stopping variable

Consider the stopping variable N of Section 3 which was defined as the first $n \geq 3$ such that $M_n \geq c_n$ where c_n was defined in (3.2). In Swanepoel and Geertsema (1976b)

TABLE 3.1

$\alpha \backslash \theta$	0,3	0,1	0,06	0,045	0,037	0,032	0,028	0,025
0,05	15,3 (1,7)	51,1 (10,2)	92,2 (24,0)	128,9 (37,9)	161,9 (51,3)	191,8 (63,9)	224,1 (77,9)	255,7 (92,0)
0,03	17,5 (1,6)	57,9 (10,3)	103,8 (24,4)	144,5 (38,7)	181,1 (52,3)	214,0 (65,2)	249,6 (79,6)	284,3 (93,9)
0,018	19,6 (1,6)	64,5 (10,4)	114,9 (24,8)	159,4 (39,2)	199,3 (53,0)	235,1 (66,1)	273,7 (80,6)	311,4 (95,1)
0,011	21,6 (1,5)	70,7 (10,5)	125,3 (25,0)	173,3 (39,6)	216,2 (53,6)	254,6 (66,6)	296,0 (81,1)	336,4 (95,8)
0,0067	23,5 (1,4)	76,7 (10,1)	135,5 (25,2)	186,9 (39,9)	232,6 (53,8)	273,7 (66,9)	317,8 (81,5)	360,7 (96,1)
0,0041	25,5 (1,5)	82,6 (10,6)	145,3 (25,3)	200,0 (40,0)	248,6 (54,0)	292,1 (67,1)	338,8 (81,6)	384,2 (96,2)
0,0025	27,4 (1,5)	88,4 (10,6)	155,0 (25,3)	212,9 (40,0)	264,3 (54,0)	310,2 (67,0)	359,5 (81,6)	407,4 (96,1)
0,0015	29,4 (1,5)	94,4 (10,7)	164,9 (25,3)	226,0 (39,9)	280,2 (53,9)	328,6 (66,9)	380,5 (81,4)	430,9 (95,9)
0,00091	31,3 (1,5)	100,1 (10,6)	174,4 (25,3)	238,7 (39,8)	295,6 (53,8)	346,4 (66,8)	400,8 (81,2)	453,6 (95,7)
0,00055	33,2 (1,5)	105,8 (10,6)	183,9 (25,2)	251,4 (39,8)	310,9 (53,6)	364,1 (66,6)	421,0 (81,0)	476,2 (95,3)

it is proved that, for $\theta > 0$, $h(N) \xrightarrow{D} X$ as $\alpha \rightarrow 0$, where

$$h(N) = -N\theta + \log \alpha^{-1} + \log_2 \alpha^{-1} + 2\theta + \log \theta^{-1} - \log 2$$

and where X has density $\exp \{-x\}$, $x > 0$. It follows that we have approximately

$$(4.1) \quad P(N > n) = 1 - 2\alpha\theta(\log \alpha^{-1})^{-1} \exp \{\theta(n - 2)\} \text{ if } h(n) > 0.$$

It may be shown that this formula satisfies (1.2) as $\alpha \rightarrow 0$. In Table 4.1 exact values of $P(N > n)$, obtained from (2.2), are compared with the approximation (4.1).

TABLE 4.1

	n	20	30	40	50	60	70	80	90	100	110
$\alpha=0,005$ $\theta=0,1$	Exact	0,9938	0,9870	0,9754	0,9540	0,9116	0,8238	0,6352	0,2196	0	0
	Approximate	0,9989	0,9969	0,9916	0,9771	0,9377	0,8305	0,5394	0	0	0
$\alpha=0,001$ $\theta=0,1$	Exact	0,9989	0,9974	0,9948	0,9899	0,9800	0,9594	0,9148	0,8164	0,5941	0,0841
	Approximate	0,9998	0,9995	0,9987	0,9965	0,9904	0,9740	0,9293	0,8079	0,4779	0

We remark that if we equate corresponding moments of $h(N)$ and X , we obtain

$$EN \doteq \theta^{-1}(\log \alpha^{-1} + \log_2 \alpha^{-1}) + 2 + \theta^{-1}(\log \theta^{-1} - \log 2 - 1)$$

which gives $EN \sim \theta^{-1} \log \alpha^{-1}$ as $\alpha \rightarrow 0$ (a fact which was rigorously established in (3.3)) and which gives additional support to (3.5).

We also find $\text{var}(N) \sim \theta^{-2}$, which has the interesting property of being independent of α .

APPENDIX

In this appendix the details of the derivations of the upper bounds of $P(N < \infty)$ in Section 2 are given.

If c_n is given by (2.2),

$$P(N < \infty) \leq c^{-1}(A_1 + A_2 + A_3)$$

where

$$A_1 = \sum_{n=20}^{\infty} 1 / n \log n (\log_2 n)^2$$

$$A_2 = 3 \sum_{n=20}^{\infty} \log_3 n / n \log n (\log_2 n)^3$$

$$A_3 = \log c \sum_{n=20}^{\infty} 1 / n \log n (\log_2 n)^3$$

Then

$$A_1 \leq \int_{19}^{\infty} dx / x \log x (\log_2 x)^2$$

$$= [-(\log_2 x)^{-1}]_{19}^{\infty}$$

$$= (\log_2 19)^{-1}$$

$$= 0,925996$$

$$\begin{aligned} A_2 &\leq 3 \sum_{n=20}^{23} \log_3 n / n \log n (\log_2 n)^3 + 3 \int_{23}^{\infty} \log_3 x dx / x \log x (\log_2 x)^3 \\ &= 0,014598 + 3[-\log_3 x / 2(\log_2 x)^2]_{23}^{\infty} + 3 \int_{23}^{\infty} dx / 2x \log x (\log_2 x)^3 \\ &= 0,014598 + 3 \log_3 23 / 2(\log_2 23)^2 + 3[-1 / 4(\log_2 x)^2]_{23}^{\infty} \\ &= 0,014598 + 3 \log_3 23 / 2(\log_2 23)^2 + 3 / 4(\log_2 23)^2 \\ &= 0,014598 + 0,153301 + 0,574289 \\ &= 0,742188 \end{aligned}$$

$$\begin{aligned} A_3 &\leq \log c \int_{19}^{\infty} dx / x \log x (\log_2 x)^3 \\ &= \log c [-1 / 2(\log_2 x)^2]_{19}^{\infty} \\ &= \log c / 2(\log_2 19)^2 \\ &= 0,428734 \log c \end{aligned}$$

so that

$$P(N < \infty) \leq c^{-1}(1,668184 + 0,428734 \log c)$$

For $c'_n = n^{-1}(\log n + \log d)$, we find

$$c_n - c_{n-1} \geq (1 - \log c - \log n) / n(n-1)$$

so that

$$y_n \leq n^{-1}(\log n + \log c - 1)$$

while

$$x_n^n = (nc)^{-1}$$

This shows

$$\begin{aligned}
P(N < \infty) &\leq c^{-1} (\sum_{n=20}^{\infty} \log n / n^2 + (\log c - 1) \sum_{n=20}^{\infty} n^{-2}) \\
&\leq c^{-1} (\int_{19}^{\infty} \log x dx / x^2 + (\log c - 1) \int_{19}^{\infty} dx / x^2) \\
&= c^{-1} ([-\log x / x]_{19}^{\infty} + \int_{19}^{\infty} dx / x^2 + (\log c - 1) \int_{19}^{\infty} dx / x^2) \\
&= c^{-1} (\log 19 / 19 + \log c / 19) \\
&= c^{-1} (0,154970 + 0,052632 \log c)
\end{aligned}$$

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